

Yu-Eop Kang

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EDUCATION

Seoul National University <i>Bachelor of Science in Aerospace Engineering</i>	Seoul, Republic of Korea August 29, 2018
Seoul National University <i>Integrated M.S./Ph.D. in Aerospace Engineering</i> Thesis: Efficient Machine Learning Framework for Extensive Design Space Exploration in Aerodynamic Optimization of Lifting Surfaces (Advisor: Kwanjung Yee)	Seoul, Republic of Korea February 26, 2025

Research Interest

Data-driven reduced order modeling

- Development of accurate and reliable surrogate model for predicting high-dimensional data.

ML/AI applications for aerodynamic shape design

- Using state-of-the-art ML/AI model for solving design challenges in aerodynamic shape optimization
- Rapid aerodynamic evaluation of airfoils and wings with large geometric variations

Uncertainty quantification

- Identification and quantification of uncertainties in numerical simulations and wind tunnel experiments

Publications

Y.E. Kang, D. Lee, K. Yee, Intuitive and feasible geometric representation of airfoil using variational autoencoder, *J. of Com. and Design Eng.*, 12(2), 2025.

Y.E. Kang*, S. Yang*, K. Yee, Physics-aware reduced-order modeling of transonic flow via β -variational autoencoder, *Physics of Fluids* 34(7), 2022. * Co-1st author

Y.E. Kang, S. Shon, K. Yee, Local non-intrusive reduced-order modeling based on soft clustering and classification algorithm, *Int. J. Num. Methods Eng.* 123(10), 2022.

Y.E. Kang, S. Min, T. Kim, K. Yee, Initial bead growth and distribution under low-speed icing condition, *Int. J. Heat and Mass Trans.* 149, 2020.

S. Shon, W.G. Ji, B. Kim, **Y.E. Kang**, K. Yee, Evaluation of snow accumulation simulation on a train using experimental results, *J. Wind Eng. Indus. Aero.* 232, 2023.

D. Lee, **Y.E. Kang**, D.H. Kim, K. Yee, Aeroelastic design and comprehensive analysis of composite rotor blades through cluster-based Kriging, *AIAA J.* 60(10), 2022.

Y. Yoon, **Y.E. Kang**, S.W. Kim, Y. Park, K. Yee, C.D. Carter, S.D. Hammack, H. Do, Proper orthogonal decomposition of continuum-dominated emission spectra for simultaneous multi-property measurements, *Energy* 254, 2022.

S. Shon, **Y.E. Kang**, Y. Hong, K. Yee, R.S. Myong, Design of hybrid airfoils for icing tunnel tests based on reduced-order modeling methods, *J. Aircr.* 59(4), 2022.

S. Kwon, **Y.E. Kang**, Y. Hong, K. Yee, Time-efficient aerodynamic load prediction using reduced-order modeling for ground noise calculation. *J. of Aerosp. Eng.*, (In press), 2025.

Oral Presentation

Y.E. Kang, D. Lee, K. Yee, Leveraging Deep Neural Networks for Efficient Prediction of Aerodynamic Performance Tables, *AIAA AVIATION 2024 Forum*, 2024.

Y.E. Kang, D. Lee, K. Yee, Physically interpretable airfoil parameterization using variational autoencoder-based generative modeling, *AIAA SCITECH 2024 Forum*, 2024.

Y.E. Kang, K. Yee, Reduced-order modeling of icing CFD data for uncertainty quantification of icing wind tunnel experiments, *SAE Technical Paper*, 2023.

Y.E. Kang, S. Min, T. Kim, K. Yee, Experimental investigation on bead growth and dispersion under low-speed icing condition, *Int. Workshop on Atmos. Icing Struct.*, 2019.

SKILLS and Techniques

Advanced proficiency in Python and MATLAB, specialized in developing ML/AI models.
Proficient in collaborative research, especially inter-disciplinary research efforts.

HONORS AND AWARDS

Outstanding Doctoral Dissertation Award **

** Highly competitive award given to one doctoral graduate per subfield (Aerodynamics, Structures, Propulsion, and Control) each year in the department of aerospace engineering at SNU.